

High-Performance 2D Rhenium Disulfide (ReS_2) Transistors and Photodetectors by Oxygen Plasma Treatment

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Transition metal dichalcogenides (TMDs) with an atomically thin 2D semiconducting structure, such as molybdenum disulfide (MoS_2), molybdenum diselenide (MoSe_2), tungsten disulfide (WS_2), and tungsten diselenide (WSe_2), have been explored as future electronic and optoelectronic devices due to their superior electrical^[1–7] and optical^[8–14] properties. When most TMD materials are scaled down to the monolayer, a transition from indirect to direct bandgap occurs due to the quantum confinement effect.^[15,16] Because of this phenomenon, most studies about optoelectronic devices based on TMDs have focused on monolayer films, where the direct bandgap property generally leads to high quantum efficiency compared to bulk-like thick TMD materials.^[17] However, despite their superior optical properties, optical device performance (e.g., photoresponsivity in a photodetector) is limited, owing to the poor absorbance resulting from the atomically thin thickness (≈ 0.7 nm).^[18,19] Research is currently underway on one of the most recently discovered next-generation TMD materials, rhenium disulfide (ReS_2). The ReS_2 maintains a direct bandgap of about 1.5 eV when transitioning from monolayer to bulk due to its very weak interlayer coupling in contrast with other TMD materials (MoS_2 , MoSe_2 , WS_2 , and WSe_2).^[20,21] This unique property may enable bulk ReS_2 to become a promising candidate for future optoelectronic devices, and it is also expected to provide an excellent responsivity and absorption efficiency. In this light, ReS_2 has been investigated as a candidate for various applications, such as thin-film transistors (TFTs),^[22–24] digital logic devices,^[23] and photodetectors.^[12,25–27] Corbet et al. and Liu et al. reported a high on/off-current ratio of 10^7 and 10^5 in the ReS_2 -based TFTs, respectively.^[22,23] In addition, Zhang et al. and

Liu et al. presented a high photoresponsivity of $88\,600\text{ A W}^{-1}$ and 16 A W^{-1} in their ReS_2 -based photodetectors, respectively.^[25,26] However, these devices were all fabricated on a monolayer or a few layers (below 5 nm) of ReS_2 , where it was very difficult to absorb sufficient light. In addition, the thickness of the used ReS_2 films was roughly adjusted through mechanical exfoliation via tape. Therefore, it is very important to properly control the thickness of ReS_2 through a new approach that is not based on mechanical cleavage and consequently achieve a sufficient thickness for the successful integration of high-performance electronic and optoelectronic ReS_2 devices. Recently, a few studies on different techniques for controlling the thickness of 2D semiconductors have been reported on MoS_2 and black phosphorus: laser thinning,^[28,29] chemical etching,^[30] and plasma etching.^[31–33]

Here, we present a simple and efficient top-down approach for implementing a high-performance TFT and photodetector on a thick ReS_2 film (above 30 nm) by controlling its thickness. To improve the electrical and optical performance, oxygen (O_2) plasma treatment was completed on the ReS_2 TFT, where we achieved a high on/off-current ratio (above 10^4), high mobility ($7.6\text{ cm}^2\text{ V}^{-1}\text{ s}^{-1}$), high photoresponsivity (above 10^7 A W^{-1}), and fast temporal response (rise time of 670 ms). This improvement originates from i) the controlled thickness of ReS_2 and ii) the intentionally created defects (traps) in the ReS_2 during the O_2 plasma treatment, which affected the drain current at the off-state and the recombination of photocarriers. First, we confirmed the ability of the O_2 plasma treatment to control the thickness on ReS_2 using optical microscopy (OM) and atomic force microscopy (AFM). To investigate the effect of O_2 plasma treatment on the ReS_2 , we then carried out X-ray photoelectron spectroscopy (XPS), Kelvin probe force microscopy (KPFM), and Raman spectroscopy analyses. Finally, the influence of the O_2 plasma process performed on the ReS_2 TFT and photodetector is discussed in terms of the electronic and optoelectronic device performance through electrical and optical measurements.

First, thick ReS_2 films were exfoliated onto the SiO_2/Si substrate by a mechanical cleavage method with blue Nitto tape, followed by O_2 plasma treatment on the ReS_2 films. The O_2 plasma treatment process was performed at 5 sccm, 470 mTorr, and 20 W and considerably low plasma power was chosen to minimize the damage on the morphology and crystal structure of the ReS_2 films. **Figure 1a–c, d–f**, respectively, show OM images and the corresponding AFM images of the as-exfoliated and O_2 -plasma-treated ReS_2 flakes. The different color contrast of ReS_2 , which is related to its thickness, was observed

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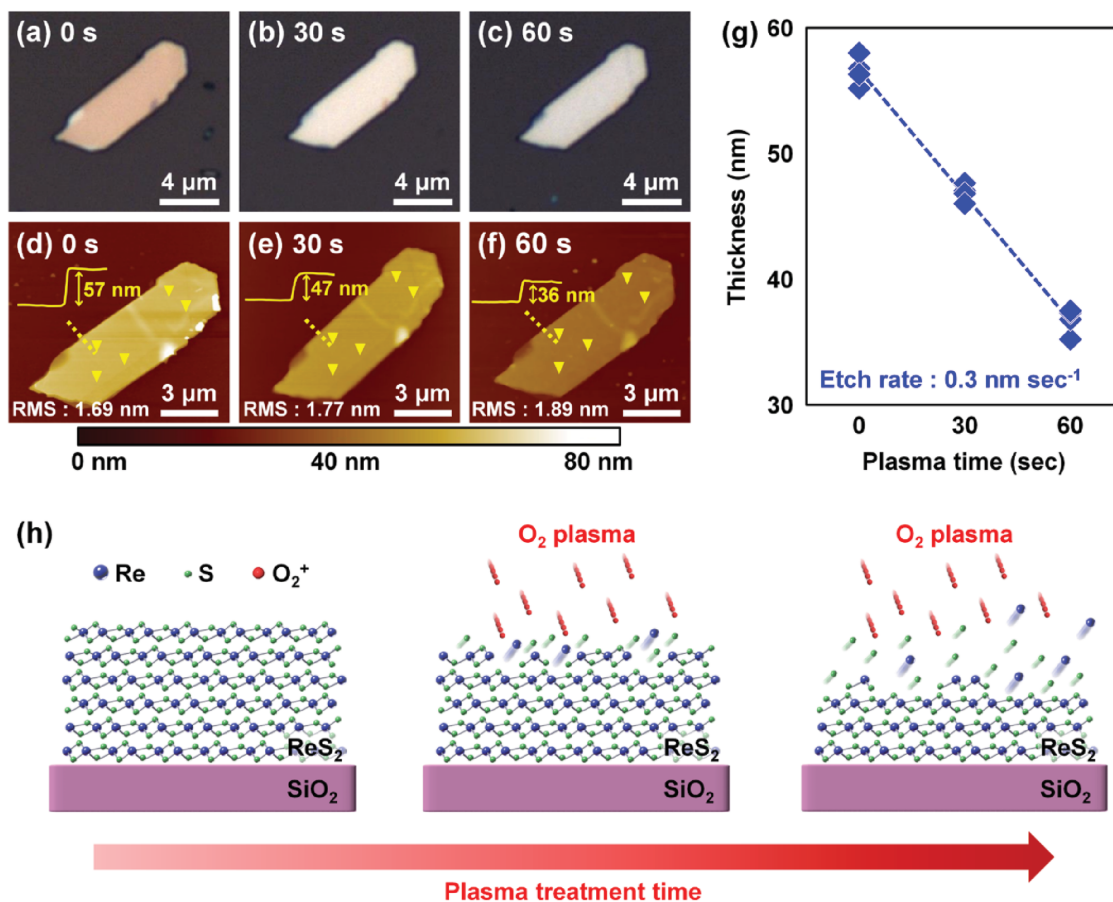


Figure 1. a–c) Optical images and d–f) corresponding AFM images of ReS₂ films: a,d) as-exfoliated, b,e) after O₂ plasma treatment for 30 s, and c,f) after O₂ plasma treatment for 60 s. g) Measured ReS₂ thicknesses on five different points of each ReS₂ film (yellow triangles in d–f) as a function of O₂ plasma treatment time. h) Schematic illustrations explaining the ReS₂ thinning mechanism by the O₂ plasma treatment.

with increasing O₂ plasma treatment time, as shown in the OM images (Figure 1a–c). The thickness of the ReS₂ film, which was confirmed through AFM analysis, was reduced from 57 to 47 nm and to 36 nm after the O₂ plasma treatment process for 0, 30, and 60 s, respectively. The root mean square (RMS) values (1.77 nm for 30 s and 1.89 nm for 60 s) of the O₂-plasma-treated ReS₂ samples seem to be almost the same as for the as-exfoliated ReS₂ sample (1.69 nm), indicating that damage and contamination issues by the O₂ plasma treatment process could be ignored. Figure 1g shows the extracted thicknesses for five different points of each ReS₂ film (yellow triangles in Figure 1d–f) as a function of O₂ plasma treatment time (0, 30, and 60 s). The thickness of the ReS₂ film seems to be uniformly diminished with increased plasma treatment time, presenting an etching speed of about 0.3 nm s⁻¹. This meant that through this O₂-plasma-based technique, it is possible to precisely adjust the thickness of the ReS₂. Figure 1h shows schematic illustrations explaining this technique in which the thickness of ReS₂ is thinned by the O₂ plasma treatment. In the other 2D semiconductors, such as MoS₂ and BP, the argon (Ar) plasma treatment process was mainly performed to control the thickness of the materials.^[31,33] The MoS₂ and BP have strong interlayer coupling energies (4.8 meV atom⁻¹ and 16 meV atom⁻¹, respectively)^[20,34] so that a high kinetic

energy is required to remove their atomic layers. In the case of Ar gas, a high kinetic energy can be obtained because it has a heavy molar mass (40 g mol⁻¹). On the contrary, due to the weak interlayer coupling energy (0.19 meV atom⁻¹) of ReS₂,^[20] the Re–S bonds can be easily broken even with a low energy. As a result, the thickness of the ReS₂ film can be controlled by the O₂ plasma treatment process, where O₂ gas has a relatively smaller kinetic energy compared to the Ar gas due to the lighter molecular mass of oxygen (32 g mol⁻¹).

In order to identify the chemical modifications and the formation of additional chemical bonds on the ReS₂ surface, XPS characterizations were performed on the ReS₂ films before and after the O₂ plasma treatment process. Figure 2a,b show the XPS spectra of Re 4f and S 2p for as-exfoliated and O₂-plasma-treated ReS₂ films. In the as-exfoliated ReS₂ film, two distinct peaks were observed at 41.68 and 44.08 eV, which correspond to the Re 4f_{7/2} and Re 4f_{5/2} states, respectively (Figure 2a). After the O₂ plasma treatment process was performed on the ReS₂ film, additional peaks were observed at 46.1 eV and 48.2 eV, which are related to ReO₃.^[35] This result indicated that the O₂ plasma treatment led to the oxidation of the ReS₂ surface by forming Re–O bonds. The calculated oxidation ratio (identified as the integrated ReO₃ peak area divided by the whole peak area) increased as the O₂ plasma treatment time increased

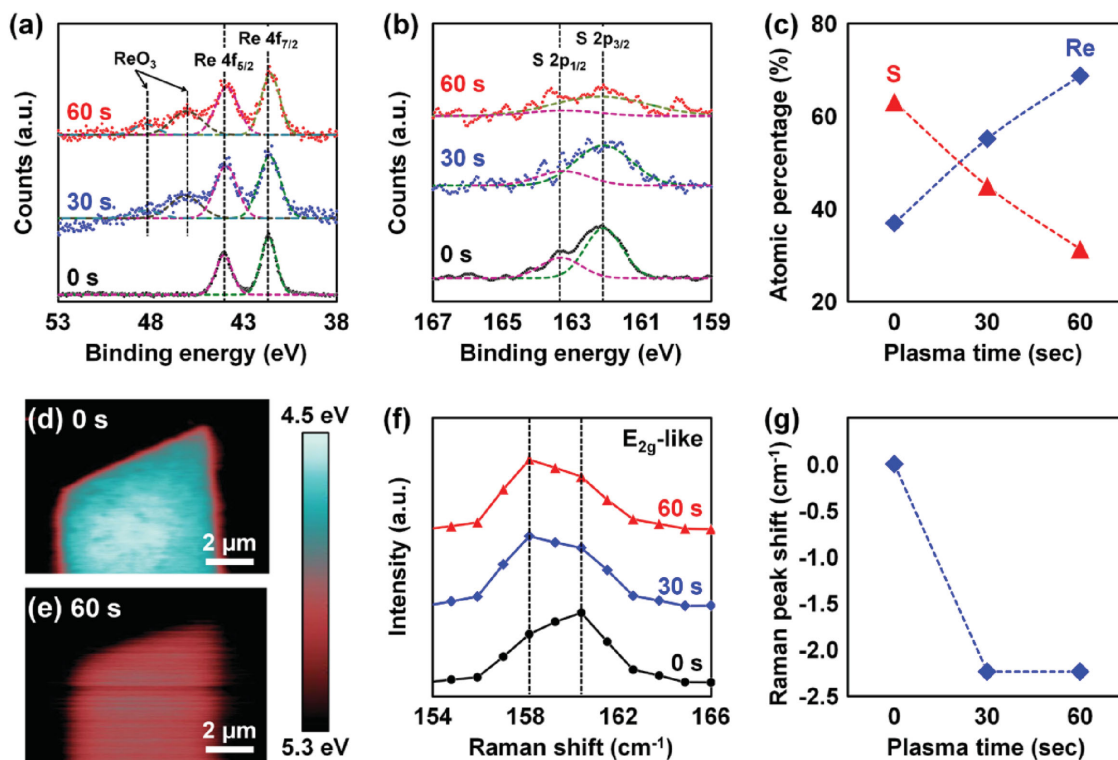


Figure 2. XPS spectra of a) Re 4f, and b) S 2p states for as-exfoliated and O₂-plasma-treated ReS₂ films. c) Relative atomic percentages of Re and S as a function of O₂ plasma treatment time, which were measured by XPS on the ReS₂ surface. Work function mapping images of the ReS₂ surface a) before and b) after O₂ plasma treatment, which were based on KPFM measurements. f) Raman spectra of as-exfoliated and O₂-plasma-treated ReS₂ films between 154 and 166 cm⁻¹. g) Extracted E_{2g}-like Raman peak shift as a function of O₂ plasma treatment time.

(32% for 30 s and 58% for 60 s), indicating further creation of Re—O bonds. In addition, after the O₂ plasma treatment, the Re 4f_{7/2} and Re 4f_{5/2} peaks were shifted toward a lower binding energy (100 meV for 30 s and 170 meV for 60 s), subsequently increasing the work function of ReS₂.^[36] Based on this behavior, the O₂ plasma treatment process is expected to introduce oxygen doping phenomenon to the ReS₂ films, where the oxygen serves as an acceptor charge.^[24] Figure 2b shows two S 2p related peaks at 162.1 eV (S 2p_{3/2}) and 163.3 eV (S 2p_{1/2}) detected in the pristine and thinned ReS₂ films, respectively. Since the S 2p peaks did not show significant changes after the O₂ plasma treatment, it seemed that S—O bonds were not formed. Furthermore, decreased intensities in the S 2p peaks were observed in the O₂-plasma-treated ReS₂ films. Because the peak intensity is related to the concentration of elements and fewer S atoms were found, it is thought that S vacancies are created by the O₂ plasma treatment process. We also extracted atomic percentages in the ReS₂ surface from the XPS data and plotted the values as a function of O₂ plasma treatment time in Figure 2c. In the as-exfoliated ReS₂ film, 36.9% Re and 63.1% S were observed, indicating that the atomic ratio of S/Re on the ReS₂ surface was approximately 1.71. By increasing the O₂ plasma treatment time, the atomic percentage of Re increased, but that of S decreased. It is predicted that the O₂ plasma treatment process i) breaks Re—S bonds, ii) creates S vacancies, and iii) forms Re—O bonds. As a result, due to the increased S vacancies by O₂ plasma treatment, a relatively high atomic percentage of Re was detected.

In order to support the oxygen doping phenomenon on the ReS₂ film, which is mentioned above in Figure 2a, we performed KPFM measurements. Figure 2d,e show the KPFM mapping images of as-exfoliated and O₂-plasma-treated ReS₂ films, respectively. To obtain the work function values of the ReS₂ films, the KPFM tip was calibrated on the highly oriented pyrolytic graphite (HOPG) surface and the contact potential difference (ΔV_{CPD}) values between the tip and the ReS₂ films were measured, which are explained in more detail in Figure S1 (Supporting Information). In the O₂-plasma-treated ReS₂ film for 60 s, an increased work function (5 eV) was obtained compared to the as-exfoliated ReS₂ film (4.6 eV), indicating that the ReS₂ surface was doped to p-type. Therefore, the O₂ plasma process was predicted to create electron-accepting sites on the surface of the ReS₂,^[24] which is in agreement with the XPS measurement results (Figure 2a). Raman analyses of the ReS₂ films before and after the O₂ plasma treatment process were also carried out to further investigate the influence of O₂ plasma treatment on ReS₂. Figure 2f shows the Raman spectra of as-exfoliated and O₂-plasma-treated ReS₂ films between 154 and 166 cm⁻¹ that correspond to the in-plane vibration of Re atoms (E_{2g}-like). The full range characterization of the Raman spectra of the ReS₂ films can be found in Figure S2 (Supporting Information). In the O₂-plasma-treated ReS₂ films, we found a red-shift phenomenon of the E_{2g}-like peak by about 2.2 cm⁻¹ as shown in Figure 2g. It is thought that the O₂ plasma process affects the ReS₂ in-plane vibrational mode and causes a p-doping phenomenon on the ReS₂. For reference, the red-shift

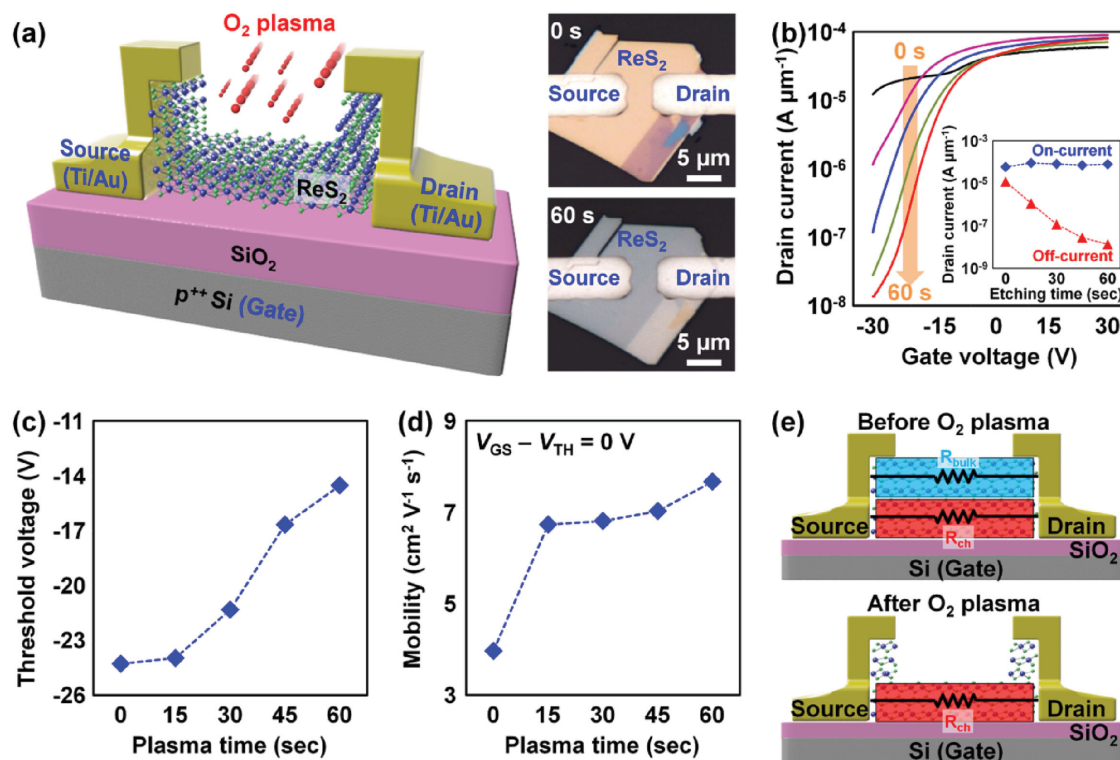


Figure 3. a) Schematic illustration of O₂-plasma-treated ReS₂ TFT and optical images before and after O₂ plasma treatment. b) I_D - V_G characteristics of ReS₂ TFTs as a function of O₂ plasma treatment time, where $V_{DS} = 5$ V. The inset shows the extracted on- (at $V_{GS} = 30$ V) and off-currents (at $V_{GS} = -30$ V) of the ReS₂ TFTs as a function of O₂ plasma treatment time. c) Threshold voltage and d) field-effect mobility as a function of O₂ plasma treatment time, which were extracted from the electrical characteristics of the ReS₂ TFTs. e) Schematic illustrations for the ReS₂ TFTs before and after O₂ plasma treatment with a resistor network model.

phenomenon of the Raman peak was reported on p-doped WSe₂.^[37] Meanwhile, the full width at half-maximum (FWHM) values of all Raman peaks in the O₂-plasma-treated ReS₂ films seem to be almost unchanged compared to the values of the pristine ReS₂ film (Figure S2, Supporting Information). Since the FWHM values are related to the crystalline quality of the ReS₂ film, it is expected to maintain its crystalline quality even after the O₂ plasma treatment process.

We then fabricated ReS₂ thin-film transistors (TFTs) and investigated the influence of the O₂ plasma treatment on ReS₂ TFTs in terms of the device performance. Figure 3a shows schematic illustration of the O₂-plasma-treated ReS₂ TFT and the optical images before and after O₂ plasma treatment. The ReS₂ flake was exfoliated onto the SiO₂/Si substrate and then Ti/Au layers were deposited onto the ReS₂ film as source/drain electrodes, where Ti was chosen to minimize the influence of contact resistance between the metal and ReS₂ channel because of its low work function (4.3 eV). We then performed electrical measurements in the ReS₂ TFTs and extracted electrical parameters (threshold voltage (ΔV_{th}) and field effect mobility (μ_{eff})) with increasing O₂ plasma treatment time. Here, we note that the O₂ plasma treatment was carried out after the source/drain electrodes were formed on the ReS₂. Figure 3b shows the drain current-gate voltage (I_D - V_G) characteristics of the ReS₂ TFT before and after the O₂ plasma treatment process (0, 15, 30, 45, and 60 s) and the inset shows the extracted on- (at $V_{GS} = 30$ V) and off-currents (at $V_{GS} = -30$ V) as a function of O₂ plasma

treatment time. As the O₂ plasma treatment time increases, the on-current seems to be unchanged, but the off-current shows a decreasing tendency. After the O₂ plasma treatment was performed on the ReS₂ TFT for 60 s, the off-current was reduced by approximately a factor of 10³, consequently presenting a much higher on/off-current ratio of 10⁴ compared to the pristine ReS₂ TFT (before the O₂ plasma treatment). For comparison, we also carried out I_D - V_G measurement on the ReS₂ TFTs after O₂ plasma treatment with higher plasma power (40 W). Here, both on- and off-currents showed a decreasing tendency as the O₂ plasma treatment time increased; this is because significant damage was probably caused to the ReS₂ channel by the high plasma power (Figure S3, Supporting Information). In addition, the V_{th} was positively shifted from -24 V to -14 V and the μ_{eff} was increased from 4 to 7.6 cm² V⁻¹ s⁻¹ by increasing the O₂ plasma treatment time from 0 s to 60 s (Figure 3c,d). Here, an on/off-current ratio of 3×10^5 and field effect mobility of 11 cm² V⁻¹ s⁻¹ were achieved in the O₂-plasma-treated ReS₂ TFT, which were comparable to the previously reported values for other ReS₂ TFTs in the literature (Figure S4, Supporting Information).^[12,22–27] For reference, similar electrical characteristics were observed in six different ReS₂ TFTs after the O₂ plasma treatment process. As shown in Figure 3e, the ReS₂ channel can be explained with two parallel resistors (channel resistance (R_{ch}), bulk resistance (R_{bulk})). R_{ch} is controlled by the gate electrode, and R_{bulk} is independent of the gate bias. For reference, the R_{bulk} in the other TMD materials, such as MoS₂ and

WSe₂, is known to have a significantly low resistivity, consequently increasing the off-current of the TFTs fabricated on the bulk TMD semiconductors.^[38,39] Therefore, the on/off-current ratio is limited in the bulk TMD TFTs because the drain current mainly flows through the bulk channel, which is not controlled by an applied gate bias. However, by applying the O₂ plasma treatment process on the ReS₂ films, the thickness of ReS₂ can be decreased, eventually suppressing the influence of the R_{bulk} in the ReS₂ TFTs. As a result, by reducing the thickness of the ReS₂, the off-current can be better controlled by an applied gate bias in the O₂-plasma-treated ReS₂ TFTs (Figure 3b). On the other hand, the on-current is mainly affected by R_{ch} regardless of ReS₂ thickness, consequently maintaining its current level even after the O₂ plasma treatment, which is discussed in more detail in Figure S5 (Supporting Information). Positively shifted V_{th} values were also obtained by increasing the O₂ plasma treatment time (Figure 3c) because the thin ReS₂ channel was depleted by a small negative gate bias so that the off-state was well controlled, even with a low negative gate bias. In addition, the gate controllability improved by reducing the thickness of ReS₂ led to an increase in the transconductance of the ReS₂ TFTs (0.76, 1.29, 1.31, 1.35, and 1.47 $\mu\text{A V}^{-1}$ after O₂ plasma treatment process for 0, 15, 30, 45, and 60 s, respectively), thereby showing an increasing tendency in μ_{FE} as more thinning of the ReS₂ thickness is applied. We note that μ_{FE} depends on the transconductance ($\partial I_{\text{DS}}/\partial V_{\text{GS}}$) at $V_{\text{GS}} - V_{\text{TH}} = 0$ because the conventional μ_{FE} extraction method was used (see the Experimental section). For reference, Das et al. previously

reported an increasing tendency in μ_{eff} by reducing the thickness of MoS₂ from 70 to 10 nm.^[40] Meanwhile, the decreased off-current and the positively shifted V_{th} can be also explained by the surface oxidation providing electron-accepting sites, which consequently causes p-type doping of the ReS₂ film. As the O₂ plasma treatment time increases from 0 to 60 s, the carrier concentration n was also reduced from 5.9×10^{12} to $6.5 \times 10^{11} \text{ cm}^{-2}$ (Figure S6, Supporting Information), which is consistent with the previous results of the XPS, KPFM, and Raman analyses (Figure 2). This decreased n resulting from the surface oxidation-based p-doping phenomenon causes the up-shift of the ReS₂ energy band, subsequently decreasing the electric field at the metal/ReS₂ junction interface. This decreased electric field at the junction is then expected to increase the effective barrier height from the metal to ReS₂. As a result, we can predict that the off-current is decreased and that V_{th} is positively shifted by the reduced electron injection probability from the metal to ReS₂, which is discussed in more detail in Figure S7 (Supporting Information). However, the p-doping by the O₂ plasma treatment does not cause a type-conversion of ReS₂ from n- to p-type.

Next, we performed optical measurements to further investigate the influence of the O₂ plasma treatment on the ReS₂ photodetectors in terms of optoelectronic device performance. Figure 4a shows a schematic illustration of the O₂-plasma-treated ReS₂ photodetector. We measured the optical characteristics of the ReS₂ photodetector, which went through the O₂ plasma treatment for 60 s, under dark and laser illuminated

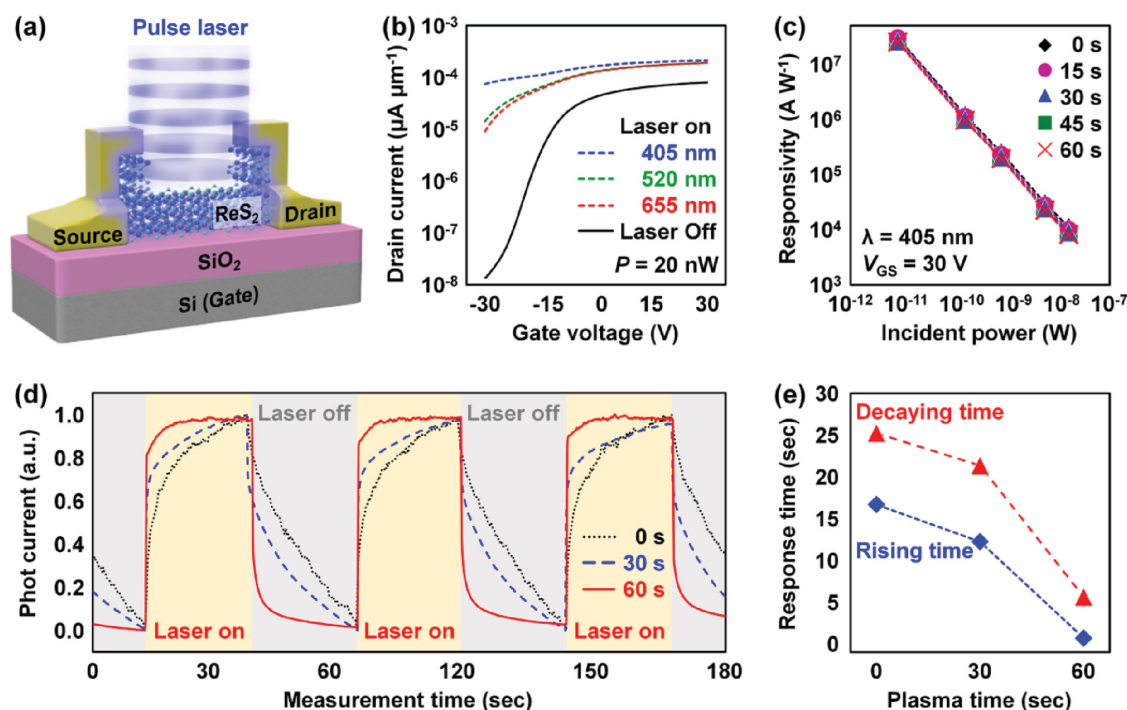


Figure 4. a) Schematic illustration of O₂-plasma-treated ReS₂ photodetector under exposure to a pulse laser. b) $I_{\text{D}}-V_{\text{G}}$ characteristics of the ReS₂ photodetector treated by O₂ plasma for 60 s under darkness and laser illuminated conditions ($\lambda = 405, 520$, and 655 nm), where the laser incident power was 20 nW . c) Photoresponsivity of the ReS₂ photodetectors as a function of incident laser power (15 nW , 5 nW , 650 pW , 130 pW , and 5 pW) according to various O₂ plasma treatment times, where $V_{\text{GS}} = 30 \text{ V}$, $V_{\text{DS}} = 5 \text{ V}$, and $\lambda = 405 \text{ nm}$. d) Normalized temporal photoresponse characteristics of as-exfoliated and O₂-plasma-treated ReS₂ photodetectors, where $V_{\text{GS}} = 0 \text{ V}$, $V_{\text{DS}} = 1 \text{ V}$. e) Rising and decaying times of the ReS₂ photodetectors as a function of O₂ plasma treatment time.

conditions ($\lambda = 405, 520$, and 655 nm), where the laser incident power was fixed at 20 nW (Figure 4b). The photocurrent seemed to have increased as the laser wavelength decreased because of the increased probability of a transition of the photocarriers by high photon energy.^[41] In order to evaluate the performance of the ReS_2 photodetectors, we extracted the photoresponsivity values from the measured photocurrents. Here, the photoresponsivity (R) was obtained by the following equation: $R = I_{\text{Photo}}/P_{\text{Light}}$, where the I_{photo} is the photocurrent ($I_{\text{Laser on}} - I_{\text{Laser off}}$) and P_{Light} is the effective incident laser power on the channel area. Figure 4c shows the extracted photoresponsivity values, where $V_{\text{GS}} = 30$ V and $\lambda = 405$ nm, as a function of an incident laser power with increasing O_2 plasma treatment time ($0, 15, 30, 45$, and 60 s). The photoresponsivity seems to be almost unchanged according to the process time because the ReS_2 film is still thick (≈ 30 nm) even after the O_2 plasma treatment process. Here, we note that an absorbance of the ReS_2 , which is dependent on the thickness of materials, seems to be saturated above 10 nm,^[20] consequently presenting a similarly high photocurrent in the ReS_2 photodetectors with a flake thicknesses above 10 nm. In particular, an increasing tendency in the photoresponsivity was observed with decreasing incident laser power because of the suppressed scattering phenomenon by the reduced number of photocarriers. In the ReS_2 photodetector treated by the O_2 plasma for 60 s, the maximum photoresponsivity was $2.5 \times 10^7 \text{ A W}^{-1}$, where $P_{\text{laser}} = 5$ pW and $\lambda = 405$ nm, which was much higher than the values of the photodetectors fabricated on other TMD materials. This comparison of the photoresponsivity with that of the other TMD-based photodetectors can be found in Table (Supporting Information). This high photoresponsivity is predicted to originate from (i) the direct bandgap property, which is independent of the ReS_2 thickness, and (ii) the high absorbance by the thick ReS_2 films (≈ 30 nm). In addition, the temporal response of the ReS_2 photodetectors was measured with increasing O_2 plasma treatment time ($0, 30$, and 60 s), where the laser on/off cycles were 30 s (Figure 4d). In Figure 4e, we plotted the rising and decaying time values of the ReS_2 photodetectors as a function of O_2 plasma treatment time, and the values were extracted between 10% and 90% of the increasing and decreasing photocurrents. In the ReS_2 photodetector before the O_2 plasma treatment, the rising and decaying time values were 16.7 s and 25.2 s, respectively. After the O_2 plasma treatment process was performed on the ReS_2 photodetectors, the temporal response was improved (rising time: 12.2 s for 30 s and 0.67 s for 60 s, and decaying time: 21.3 s for 30 s and 5.6 s for 60 s). This improvement seems to be attributed to the created defects (traps) on the surface of the ReS_2 during the O_2 plasma treatment. The defect (or trap) density (D_T) on the surface of ReS_2 is predicted to increase from 4.3×10^{11} to $1.3 \times 10^{12} \text{ cm}^{-2}$ ($\Delta D_T = 8.6 \times 10^{11} \text{ cm}^{-2}$) after the O_2 plasma treatment (Figure S8, Supporting Information). The long lifetime of the photocarriers normally degrades the temporal response characteristics because photocurrents are slowly increased or decreased due to the residual photocarriers in the ReS_2 channel when the laser turns on or off, respectively. However, if the trap states are formed in the bandgap of ReS_2 (surface region) by the O_2 plasma treatment, it would increase the recombination rate of the photocarriers, consequently reducing the photocarrier

lifetime. As a result, the O_2 -plasma-treated ReS_2 photodetectors had lots of surface defects (or traps) and much faster temporal response is expected as a result.

In conclusion, we demonstrated high-performance ReS_2 -based TFT and photodetector with high on/off-current ratio (10^4), high mobility ($7.6 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$), high photoresponsivity ($2.5 \times 10^7 \text{ A W}^{-1}$), and fast temporal response (rising and decaying time of 670 ms and 5.6 s, respectively) through O_2 plasma treatment. This high-performance originates from (i) the controlled thickness of ReS_2 and (ii) the created defects (traps) on the surface of the ReS_2 during the O_2 plasma treatment, the former of which affects the drain current at the off-state and the latter impacts the recombination of photocarriers. We confirmed the controllability of the thickness by O_2 plasma treatment on ReS_2 by using OM and AFM. The thickness of the ReS_2 film was uniformly thinned with increasing O_2 plasma treatment, presenting an etching speed of about 0.3 nm s^{-1} . To investigate the effect of O_2 plasma treatment on the ReS_2 , we then carried out XPS, KPFM, and Raman analyses. The ReS_2 surface seemed to be oxidized by forming $\text{Re}-\text{O}$ bonds and electron-accepting sites were expected to be created on the ReS_2 surface by the O_2 plasma treatment. The influence of the O_2 plasma process on the ReS_2 TFT and photodetector was also discussed in terms of the electronic and optoelectronic device performance through electrical and optical measurements. As the O_2 plasma treatment time increased, we observed improved electrical and optoelectrical properties (decreased off-current, increased on/off-current ratio, positively shifted threshold voltage, increased field effect mobility, and reduced rising/decaying times) because the ReS_2 film was sufficiently thinned and lots of surface defects (or traps) were created by the O_2 plasma treatment. In addition, very high photoresponsivity was achieved in the O_2 -plasma-treated ReS_2 photodetector, which was even higher than the values of the other TMD-based photodetectors. Our study suggests that bulk ReS_2 has great potential and may be a promising candidate for future TMD-based 2D electronic and optoelectronic applications.

Experimental Section

Fabrication of ReS_2 Device: ReS_2 flakes were exfoliated onto a 90 nm thick SiO_2 layer on a heavily doped p-type Si substrate by adhesive tape (224SPV, Nitto). To remove tape residues, the ReS_2 samples were rinsed with acetone for 1 h. For the fabrication of back-gated ReS_2 transistors, source and drain electrode regions (channel length and width were $5 \mu\text{m}$) were patterned by optical lithography and then Ti (10 nm)/Au (30 nm) layers were deposited in an electron-beam evaporating system, followed by a lift-off process.

Oxygen (O_2) Plasma Process on ReS_2 : The O_2 plasma treatment process was carried out on the ReS_2 flakes and devices by a plasma machine (Miniplasma Cube, PLASMAART). To stabilize the chamber conditions, O_2 gas flowed for 3 min before the O_2 plasma treatment. The following conditions, reactive ion etcher power (20 W), plasma pressure (470 mTorr), and O_2 flow rates (5 sccm), were used during the oxygen plasma treatment process to control the thickness of the ReS_2 films.

Characterization of Plasma-Treated ReS_2 : ReS_2 samples were analyzed by Raman spectroscopy (XperRam200 compact, NANOBASE), XPS (ESCA200, VG Microtech Inc.), AFM (SPA-300, Seiko instrument), and KPFM (NTEGRA Spectra, NT-MDT) before and after the oxygen plasma treatment. The Raman spectroscopy with an excitation wavelength of

532 nm was used, the laser beam size was 1 μm , and the instrumental spectral resolution was about 1 cm^{-1} . An integration time of 5 s and a spectrometer with 1800 grooves per mm were used for the test. The XPS measurements used a Mg K α twin-anode source and the X-ray incident angle was 0°.

Electrical Characterization of Plasma-Treated ReS₂ Devices: The ReS₂ back-gated transistors were characterized by using a B2912A semiconductor parameter analyzer under ambient conditions. We then extracted the threshold voltage (V_{th}) and field-effect mobility (μ_{FE}) from the measured $I_{\text{D}}-V_{\text{G}}$ data, where all drain currents (I_{D}) were normalized by the channel width (W). The μ_{FE} was extracted from $\mu_{\text{FE}} = L/(WV_{\text{D}}C_{\text{OX}}) \times (\partial I_{\text{D}}/\partial V_{\text{G}})$, where L and W are the length and width of the channel, respectively, and C_{OX} is $\epsilon_{\text{OX}} \times \epsilon_0/t_{\text{OX}}$, which is the gate oxide capacitance per unit area.

Optical Characterization of Plasma-Treated ReS₂ Devices: To investigate ReS₂ photodetector devices, we performed the electrical measurements ($I_{\text{D}}-V_{\text{G}}$) under both dark and illuminated conditions. Wavelengths of the dot laser source were 405, 520, and 655 nm, where its optical power was 15 nW. By using the measured $I_{\text{D}}-V_{\text{G}}$ data, we calculated the photoresponsivity (R) from $R = I_{\text{Photo}}/P_{\text{Light}}$, where I_{Photo} is the generated photocurrent ($I_{\text{Laser on}} - I_{\text{Laser off}}$) and P_{Light} is the effective incident optical power on the channel area. Both $I_{\text{Laser on}}$ and $I_{\text{Laser off}}$ were extracted under $V_{\text{GS}} = 30$ V. The variable P_{Light} was defined as $P_{\text{Total Light}} \times (A_{\text{Channel}}/A_{\text{Laser}})$, where $P_{\text{Total Light}}$ is the total light intensity, A_{Channel} is the effective channel area of the device, and A_{Laser} is the illuminated area. The effective incident power of the laser was adjusted to 15 nW, 5 nW, 650 pW, 130 pW, and 5 pW. By turning on and off with a cycle of 30 s, the photo-switching properties of the ReS₂ photodetectors were also investigated in terms of rising and decaying times, which were extracted between 10% and 90% of increasing and decreasing photocurrents.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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